Sam Laing

ML Research Engineer | LLM Optimization

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Personal Statement

Research Engineer at ELLIS/OpenEuroLLM specializing in large-scale multilingual LLM pretraining, scaling laws, and optimization, with experience running Megatron-LM experiments across 1,024+ H100 GPUs. My research focuses on matrix-aware optimizers and training efficiency. I hold a First Class mathematics degree (Gold Medal) from Trinity College Dublin and an MSc in Machine Learning from Tübingen (1.3), complemented by two years building and deploying production ML systems in fintech at SoftCo.

Professional Skills

Systems and Languages: Python, Triton, CUDA, SQL (DuckDB/PostgreSQL), Linux, SLURM, Docker

ML Frameworks & Scaling: PyTorch, MegatronLM, Torch Titan, JAX, WandB, Torch Lightning, Optuna

Research & Theory: LLM Pretraining Optimization (Muon, AdamW), Distributed Training (TP/DP/FSDP), Trustworthy ML, Bayesian Inference, Time Series

Experience

OpenEuroLLM/ELLIS — Tübingen, Germany
Research Engineer — *April 2026 – Present*

- Currently leading a project benchmarking the Muon optimizer against AdamW at scale, aiming to improve pretraining compute efficiency for the consortium's model series.
- Ran a large-scale pretraining sweep on Megatron-LM across **1024+ NVIDIA H100s**, debugging distributed training failures (NCCL hangs, OOMs, gradient sync issues) to keep multi-day jobs stable at scale.
- Fitted scaling laws from sweep results to inform compute-optimal model/data sizing decisions for **up to 70B parameter** multilingual LLMs.
- Built internal tooling to monitor training runs in real time (loss curves, gradient norms, throughput), cutting time-to-detect for failed or degrading runs.

ELLIS Institute/Max Planck Institute — Tübingen, Germany
Research Intern — *Oct 2025 – March 2026*

- Researched optimization dynamics in GPT-style language model pretraining, investigating the Muon optimizer's advantages over AdamW at small and large scale (Preprint).

SoftCo — Dublin, Ireland
Machine Learning Engineer — *April 2023 – April 2025*

- Developed and deployed the core Machine Learning IP that served as a primary value driver for the company's **€100M+ acquisition** by Keensight Capital.
- Engineered a Bayesian inference module and multi-label classifier achieving **95%+ touchless processing** in production — end-to-end ownership from modeling to deployment.
- Built ETL pipelines and data parsers to automate preprocessing, reducing model training cycle latency and scaling data ingestion to production volumes.
- Built time-series forecasting models that directly optimized cloud compute allocation and reduced operational costs.

Education

University of Tuebingen — Germany
MSc. in Machine Learning; ***Grade 1.3 (sehr gut)*** — *Oct 2022 – July 2025*

- **Master's Thesis:** Fundamental research into adaptive and geometry-aware optimizers (in particular, Muon) for transformer-based pretraining efficiency and understanding training dynamics. Supervised by Dr. Antonio Orvieto

Trinity College Dublin — Ireland
BA Mathematics; ***First Class Honors with Gold Medal*** — *Sep 2018 – Jun 2022*

- **Bachelor Thesis:** Applied category theoretical techniques to prove several fundamental theorems of algebraic topology.

Awards & Achievements

Trinity College Academic Gold Medal: Awarded for academic excellence throughout four years of study